

Lecture 21

Bayesian GLM and random effects models

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(Edited from the lecture notes by Brian J. Reich, NC State)



Generalized linear models

- ▶ Other forms of regression follow naturally from linear regression.
- ▶ For example, for binary responses $Y_i \in \{0, 1\}$ we might use logistic regression

$$\text{logit}[\text{Prob}(Y_i = 1)] = \eta_i = \beta_0 + \beta_1 X_{i1} + \dots + \beta_p X_{ip}.$$

- ▶ The logit link is the log-odds $\text{logit}(x) = \log[x/(1 - x)]$.
- ▶ Then β_j represents the increase in the log odds of an event corresponding to a one-unit increase in covariate j .
- ▶ The expit transformation $\text{expit}(x) = \exp(x)/[1 + \exp(x)]$ is the inverse, and

$$\text{Prob}(Y_i = 1) = \text{expit}(\eta_i) \in [0, 1].$$

Bayesian logistic regression

- ▶ Bayesian logistic regression requires a prior for β .
- ▶ All of the prior we have discussed for linear regression (Zellner, BLASSO, etc) are also applicable here.
- ▶ Computationally the full conditional distributions are no longer conjugate and so we must use Metropolis sampling.
- ▶ The R function `MCMClogit` does this efficiently.
- ▶ Other GLMs (e.g., Poisson regression, probit regression) are similar to implement using Bayesian methods.

Probit and Poisson regression

- ▶ For binary responses $Y_i \in \{0, 1\}$, we might use probit regression

$$\Phi^{-1}[\text{Prob}(Y_i = 1)] = \eta_i = \beta_0 + \beta_1 X_{i1} + \dots + \beta_p X_{ip}.$$

- ▶ The probit link is $\text{probit}(x) = \Phi^{-1}(x)$, where Φ is Normal(0, 1) CDF.
- ▶ For count responses $Y_i \in \{0, 1, \dots\}$, we might use $Y_i \overset{\text{Indep}}{\sim} \text{Poisson}(\lambda_i)$ where

$$\log[\lambda_i] = \eta_i = \beta_0 + \beta_1 X_{i1} + \dots + \beta_p X_{ip}.$$

- ▶ Computationally the full conditional distributions are no longer conjugate and so we must use Metropolis sampling.

Random effects

- ▶ Linear regression assumes that the errors are independent.
- ▶ This is invalid if data are grouped.
- ▶ For example, n classrooms each have m students.
- ▶ It might be reasonable to assume the classrooms are independent, but the students within a class are likely dependent.
- ▶ Choosing random effects is a natural way to account for this dependence.

One-way random effects model

- ▶ Say Y_{ij} is the score for student i in class j .
- ▶ The random effects model is

$$Y_{ij} = \alpha_j + \varepsilon_{ij}.$$

- ▶ The random effect for classroom j is α_j .
- ▶ This is viewed as a random draw from the population,

$$\alpha_j \stackrel{IID}{\sim} \text{Normal}(\mu, \tau^2).$$

- ▶ The population is described by μ and τ .
- ▶ The random errors are $\varepsilon_{ij} \stackrel{IID}{\sim} \text{Normal}(0, \sigma^2)$, independent over i and j .

One-way random effects model

- ▶ Conditioned on the classroom mean α_j all observations are independent:

$$Y_{ij} | \alpha_j \overset{Indep}{\sim} \text{Normal}(\alpha_j, \sigma^2).$$

- ▶ Marginalizing over the random effects gives

$$\text{Cor}(Y_{ij}, Y_{uv}) = \begin{cases} 0 & \text{for } j \neq v \\ \frac{\tau^2}{\sigma^2 + \tau^2} & \text{for } j = v \end{cases}$$

- ▶ Therefore, in this model observations with the same classroom are correlated.

One-way random effects model

- ▶ To complete the Bayesian model, we must specify priors for μ , σ^2 and τ .
- ▶ A normal prior with large variance for μ is fine.
- ▶ Improper priors must be used cautiously for complicated models.
- ▶ A natural prior for the variances is

$$\tau^2, \sigma^2 \stackrel{IID}{\sim} \text{Inverse-Gamma}(a, b).$$

- ▶ All full conditional distribution are conjugate and MCMC sampling is very fast.

One-way random effects model

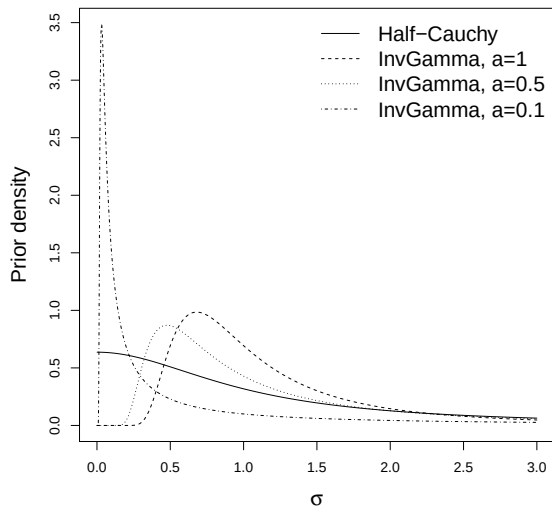
- ▶ However, under the inverse gamma prior for the variances the induced priors for σ and τ have no mass at zero.
- ▶ Gelman recommends the half-Cauchy prior for the SD

$$p(\sigma) = \frac{2}{\pi(1 + \sigma^2)},$$

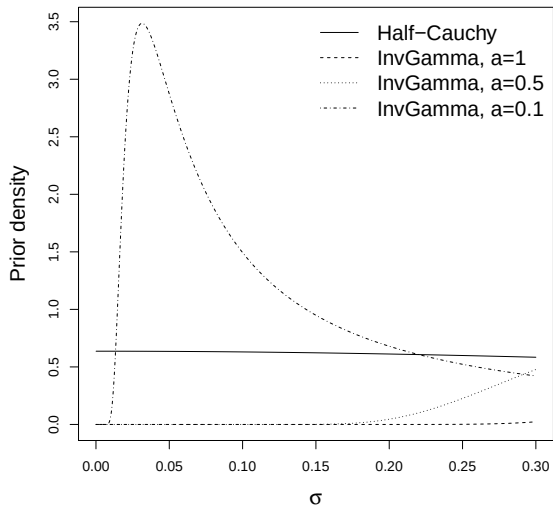
i.e., a Student-t density with 1 df restricted to be positive.

- ▶ This PDF is flat around zero and has heavy tails.
- ▶ This is very easy to code in JAGS.
- ▶ For large sample these give similar results, but half-Cauchy is often preferred.

Prior for standard deviation



Prior for standard deviation (zoomed in around 0)



Confusion about random effects

- ▶ MCMC does not distinguish between random effects and other parameters.
- ▶ For example, σ , τ , μ and α_j are all treated as random in a Bayesian analysis.
- ▶ However, α_j is called a “random” effect because it represents a random draw from the fixed $\text{Normal}(\mu, \tau^2)$ population of classroom means.
- ▶ Try to implement Gibbs sampling for this model!

Linear mixed models

- ▶ Consider the model

$$Y_{ij} = \beta_0 + X_{ij}\beta_1 + \alpha_j + \varepsilon_{ij},$$

where X_{ij} is the age of student i in class j .

- ▶ The regression coefficients β_0 and β_1 apply to all students are all called “fixed effects”.
- ▶ The random effect is $\alpha_j \sim \text{Normal}(0, \tau^2)$.
- ▶ A linear model with both fixed and random effects is called a **linear mixed model**.

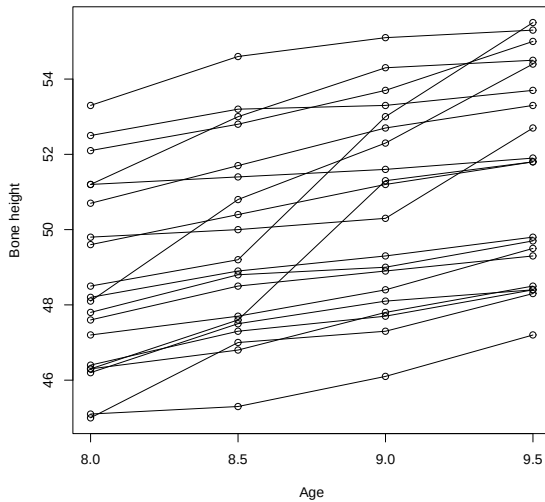
Random slopes model

- ▶ Let Y_{ij} be the j^{th} observation for subject i .
- ▶ As an example, consider the data plotted on the next slide were Y_{ij} is the bone height for child i at age X_j .
- ▶ Here we might specify a different regression for each child to capture variability over the population of children:

$$Y_{ij} \sim \text{Normal}(\gamma_{0i} + X_j\gamma_{1i}, \sigma^2).$$

- ▶ $\gamma_i = (\gamma_{i0}, \gamma_{i1})^T$ controls the growth curve for child i .
- ▶ These separate regression are tied together in the prior, $\gamma_i \sim \text{Normal}(\beta, \Sigma)$, which borrows strength across children.
- ▶ This is a linear mixed model: γ_i are random effects specific to one child and β are fixed effects common to all children.

Bone height data



Prior for a covariance matrix

- ▶ The random-effects covariance matrix is $\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{bmatrix}$.
- ▶ σ_1^2 is the variance of the intercepts across children.
- ▶ σ_2^2 is the variance of the slopes across children.
- ▶ σ_{12} is the covariance between the intercepts and slopes.
- ▶ Prior 1: $\sigma_1^2, \sigma_2^2 \sim \text{Inverse-Gamma}$ and $\rho = \frac{\sigma_{12}}{\sigma_1 \sigma_2} \sim \text{Unif}(-1, 1)$.
- ▶ Prior 2: Inverse Wishart works better in higher dimensions.

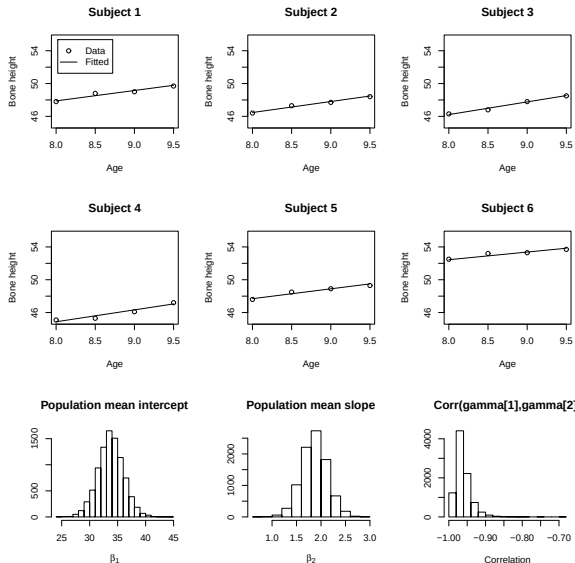
Inverse Wishart distribution

- ▶ The inverse Wishart distribution is the most common prior for a $p \times p$ covariance matrix.
- ▶ It reduces to the inverse gamma distribution if $p = 1$.
- ▶ Say $\Sigma \sim \text{Inverse-Wishart}(\kappa, R)$ where $\kappa > p + 1$ and R is a $p \times p$ covariance matrix. Both κ and R are hyperparameters.
- ▶ The PDF is
$$f(\Sigma) \propto |\Sigma|^{-(\kappa+p+1)/2} \exp \left[-\frac{1}{2} \text{trace}(R\Sigma^{-1}) \right].$$
- ▶ The mean is $\frac{1}{\kappa-p-1} R$.

Full conditional distributions

- ▶ The hierarchical model is:
 - ▶ $Y_{ij} \sim \text{Normal}(\gamma_{0i} + X_i \gamma_{1i}, \sigma^2)$
 - ▶ $\gamma_i \sim \text{Normal}(\beta, \Sigma)$
 - ▶ $p(\beta) \propto 1$
 - ▶ $\sigma^2 \sim \text{Inverse-Gamma}(a, b)$
 - ▶ $\Sigma \sim \text{Inverse-Wishart}(\kappa, R)$
- ▶ The full conditionals are all in closed forms (check!).
- ▶ Try to write the JAGS code!

Bone height data - fitted values



Linear models with correlated errors

- ▶ An alternative to using random effects to capture dependence is to model correlation directly.
- ▶ For example, say the observations are collected at n different spatial locations.
- ▶ Denote the measurement at lat/lon s_i as Y_i .

- ▶ We might fit the model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i,$$

where the residual errors ε_i have spatial correlation.

- ▶ A common model is

$$\text{Cov}(\varepsilon_i, \varepsilon_j) = \sigma^2 \exp(-d_{ij}/\phi).$$

- ▶ The parameter ϕ controls the exponential decay of the correlation as distance between sites, d_{ij} , increases.

Linear models with correlated errors

- ▶ This is straightforward (though often slow) to fit using MCMC.

- ▶ The likelihood is multivariate normal

$$\mathbf{Y}|\boldsymbol{\beta}, \sigma^2, \rho \sim \text{Normal}\left(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \boldsymbol{\Sigma}(\phi)\right).$$

- ▶ The $n \times n$ correlation matrix $\boldsymbol{\Sigma}(\phi)$ has (i, j) element $\exp(-d_{ij}/\phi)$.

- ▶ This last piece is to set a prior for ϕ .

- ▶ A uniform prior between 0 and the maximum distance between points is an option.

- ▶ This type of modeling is also useful for time series data.

Thank you!